

Supplementary Materials

Transfer learning from computed stability data for nanobody melting-temperature prediction

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Supplementary methods

Details of the heterogeneous MD data set

The heterogeneous data set was assembled from a SABDab summary and the corresponding structures [1]. For each PDB identifier, we selected the first heavy-chain entry whose chain was present in the structure file. After requiring a usable sequence, topology, trajectory, and hydrophilic native-contact label, 763 samples remained. Some sequences occurred in more than one PDB entry; native-contact values were calculated separately for each structure and sampled individually during training, duplicate sequences were not collapsed, and the model did not receive the PDB identifier. Sequences exceeding the 158-residue model input were truncated at the end.

Structures were prepared and protonated at pH 7.4 with PDBFixer, pdb2pqr, and PROPKA [2, 3]. Each protein was solvated in OPC water with 15 Å padding and 0.15 M NaCl and described with ff19SB [4, 5]. Equilibration at 300 K used 1 ns NVT and 1 ns NPT with a 2-fs timestep, after which two 1-ns restrained NVT stages at 400 K preceded 100 ns of unrestrained 400-K NVT production. Production used hydrogen-mass repartitioning to 4 amu, a 4-fs timestep, and output every 100 ps. OpenMM used PME electrostatics with a 1.0-nm cutoff, constraints on bonds to hydrogen, and a Langevin middle integrator with 1 ps⁻¹ friction [6].

The contact calculation used backbone heavy-atom pairs separated by more than three residues and within 0.45 nm in the reference structure, counting a contact only when at least one residue was hydrophilic, as for the single mutation scans. Each structure was referenced to its own parent equilibration structure, and Q was averaged over the [10, 40) ns production window sampled every 100 ps (300 frames). The two MD data sets thus shared one Q definition, contact set, native reference, and averaging window, and differed only in the sequences they covered.

FEP charge-change check

For charge-changing mutations in a cubic 70-Å box and a solvent dielectric constant of 78, we tested the leading analytical periodic net-charge (Ewald self-energy) correction [7], $0.086(\Delta q)^2 \text{ kcal mol}^{-1}$. We applied it and repeated the sign reversal, robust scaling, Yeo–Johnson transform with standardization, and min–max scaling used for the training labels. Across the 284 charge-changing mutations the scaled labels barely moved, with a maximum shift of 0.008 on the [0, 1] scale and a correlation of 0.99989 with the uncorrected labels, so the training labels use the uncorrected $\Delta\Delta G$.

Model-size checks

We fine-tuned the 35M- and 650M-parameter ESM2 encoders without size-specific hyperparameter searches. For both sizes, the Tm-only model used the shared architecture, dropout 0.05, and an encoder learning rate of 1×10^{-4} . The FEP model used the shared architecture with separate 1MEL and 4IDL heads, $n = 320$ samples per structure, dropout 0.15, and an encoder learning rate of 3×10^{-5} . Both conditions used a non-encoder learning rate of 3×10^{-4} , weight decay 0.04, and ten training seeds.

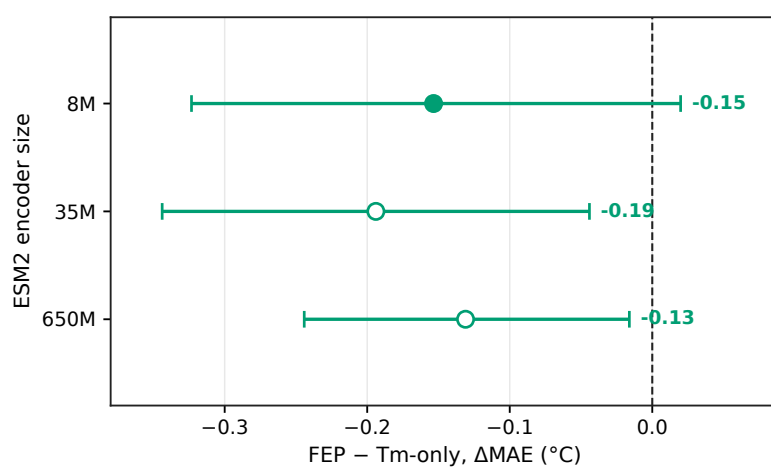


Figure S1 The FEP result held across ESM2 encoder sizes. FEP-minus-Tm-only MAE (ΔMAE) with a fine-tuned encoder; negative values mean lower Tm error with FEP. The 8M point is the five-seed model selected by the two-stage search, and the 35M and 650M points are ten-seed checks without size-specific tuning. Whiskers are 95% paired bootstrap intervals over the 396 Tm test sequences.

References

- [1] Dunbar, J., Krawczyk, K., Leem, J., Baker, T., Fuchs, A., Georges, G., et al. SAbDab: the structural antibody database. *Nucleic Acids Research* 42, D1140–D1146 (2014). <https://doi.org/10.1093/nar/gkt1043>
- [2] Dolinsky, T. J., Nielsen, J. E., McCammon, J. A., Baker, N. A. PDB2PQR: an automated pipeline for the setup of Poisson–Boltzmann electrostatics calculations. *Nucleic Acids Research* 32, W665–W667 (2004). <https://doi.org/10.1093/nar/gkh381>
- [3] Olsson, M. H. M., Søndergaard, C. R., Rostkowski, M., Jensen, J. H. PROPKA3: consistent treatment of internal and surface residues in empirical pK_a predictions. *Journal of Chemical Theory and Computation* 7, 525–537 (2011). <https://doi.org/10.1021/ct100578z>
- [4] Tian, C., Kasavajhala, K., Belfon, K. A. A., Raguet, L., Huang, H., Miguels, A. N., et al. ff19SB: amino-acid-specific protein backbone parameters trained against quantum mechanics energy surfaces in solution. *Journal of Chemical Theory and Computation* 16, 528–552 (2020). <https://doi.org/10.1021/acs.jctc.9b00591>
- [5] Izadi, S., Anandakrishnan, R., Onufriev, A. V. Building water models: a different approach. *The Journal of Physical Chemistry Letters* 5, 3863–3871 (2014). <https://doi.org/10.1021/jz501780a>
- [6] Eastman, P., Swails, J., Chodera, J. D., McGibbon, R. T., Zhao, Y., Beauchamp, K. A., et al. OpenMM 7: rapid development of high performance algorithms for molecular dynamics. *PLOS Computational Biology* 13, e1005659 (2017). <https://doi.org/10.1371/journal.pcbi.1005659>
- [7] Rocklin, G. J., Mobley, D. L., Dill, K. A., Hünenberger, P. H. Calculating the binding free energies of charged species based on explicit-solvent simulations employing lattice-sum methods: An accurate correction scheme for electrostatic finite-size effects. *The Journal of Chemical Physics* 139, 184103 (2013). <https://doi.org/10.1063/1.4826261>